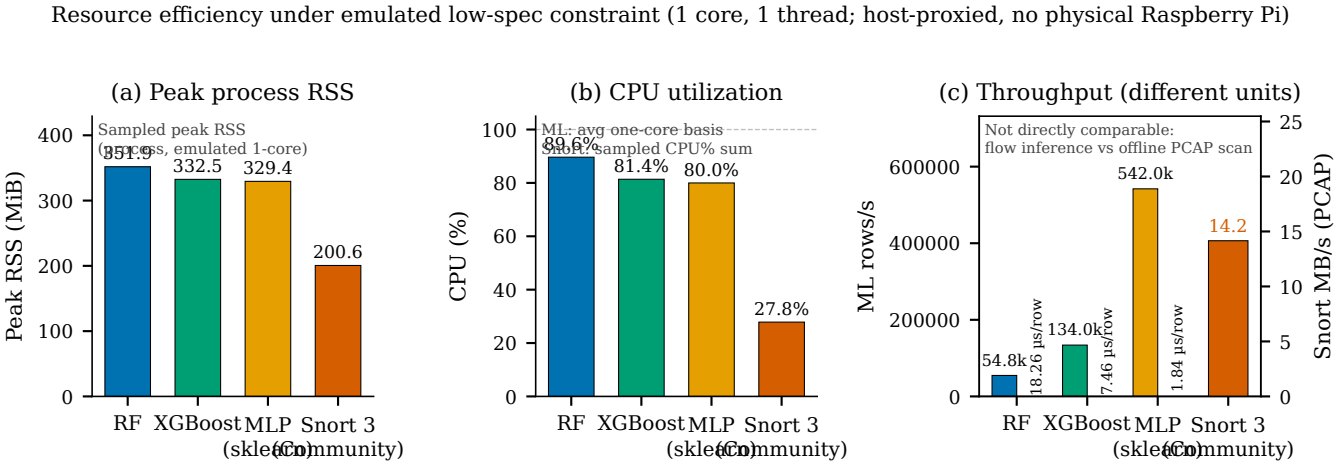
Resource efficiency under emulated low-spec constraint (1 core, 1 thread; host-proxied, no physical Raspberry Pi)
(a) Peak process RSS
Sampled peak RSS
(process, emulated 1-core)
Peak RSS (MiB)
400
300
200
100
0
351.9
332.5
329.4
200.6
RF
XGBoost
MLP (sklearn)
Snort 3 (Community)
(b) CPU utilization
ML: avg one-core basis
Snort: sampled CPU% sum
CPU (%)
100
80
60
40
20
0
89.6%
81.4%
80.0%
27.8%
RF
XGBoost
MLP (sklearn)
Snort 3 (Community)
(c) Throughput (different units)
Not directly comparable:
flow inference vs offline PCAP scan
ML rows/s
600000
400000
200000
0
Snort MB/s (PCAP)
25
20
15
10
5
0
54.8k
18.26 µs/row
134.0k
7.46 µs/row
542.0k
1.84 µs/row
14.2
RF
XGBoost
MLP (sklearn)
Snort 3 (Community)